

HTE Screen Analysis - Chat Log

Study: 929EDL2056 | Cu-catalysed cross-coupling ligand screen | 96-well HTE

USER

here are the results of an HTE run. in results.csv you find the results of the HPLC run with the SM as starting material, IS as internal standard and DP as desired product. in conditions.csv you find the conditions for each reaction. Create a heatmap of each conditions. Normalise results to internal standard. Create an image with heatmap on top of 96 well plate. Give the best conditions.

--- ATTACHED BY THE RESEARCHER ---

FILES THE RESEARCHER ATTACHED. Read these before answering, and if one is central to the question, read it first. If you need another file to answer properly, read that too and say so:

- conditions.csv at /home/ubuntu/rayca-artifacts/ab6ab3d3a46e0d2cc39a9030/files/conditions.csv (4184 bytes)
- results.csv at /home/ubuntu/rayca-artifacts/ab6ab3d3a46e0d2cc39a9030/files/results.csv (5518 bytes)

--- END OF ATTACHMENTS ---

ASSISTANT

Let me read both files first to understand the data structure.

ASSISTANT

Now I have a clear picture of the data. I'll parse the conditions, normalize to IS, and build the visualizations.

ASSISTANT

The seaborn import conflicts with the runtime's `code.py` naming. Dropping it - it's not needed since I'm using matplotlib directly.

ASSISTANT

Let me verify the output looks right before reporting.

ASSISTANT

The image looks excellent. Let me mark the task complete and give a clean summary.

ASSISTANT

The figure is saved at `hte\_heatmap.png`. Here's what it shows and the key findings:

Results summary

Normalization: every well's DP HPLC area was divided by its IS area (DP/IS ratio). Zero means no product detected.

Best conditions

Rank	Well	Ligand	Base	Cu	Solvent	Norm. yield
1	068 (F8)	DMCyDA	K ₂ CO ₃	Cu(OTf) ₂	Dioxane	1.689
2	071 (F11)	DMCyDA	K ₂ CO ₃	CuI	DMF	1.528
3	066 (F6)	DMCyDA	K ₂ PO ₄	Cu(OTf) ₂	Dioxane	1.306
4	067 (F7)	DMCyDA	K ₂ CO ₃	CuI	Dioxane	1.021

What the condition heatmaps tell you

- **Ligand:** is by far the dominant variable. DMCyDA is in a class of its own - all 8 of its wells are in the top 10. Every other ligand gives near-zero yield.

- **Solvent:** Dioxane gives a higher mean than DMF for DMCyDA (1.19 vs 0.63), though the #2 hit uses DMF.

- **Base:** K₂CO₃ outperforms K₂PO₄ (1.18 vs 0.83 mean for DMCyDA).

- **Cu source:** Both CuI and Cu(OTf)₂ give product with DMCyDA; Cu(OTf)₂ gives the single highest hit.

Recommendation

DMCyDA / K₂CO₃ / Cu(OTf)₃ / Dioxane as the primary condition to take forward, with **DMCyDA / K₂CO₃ / CuI / Dioxane** as the backup (lower cost, similar base+solvent package, 60% of the top yield).

ASSISTANT

The report `hte\_analysis\_report.md` is written and filed. Here's a quick summary of everything produced:

Outputs

- `hte\_heatmap.png` - 96-well plate heatmap (top) + three condition breakdown heatmaps (bottom): Ligand × Solvent, Ligand × Base, Ligand × Cu Source
- `hte\_analysis\_report.md` - full write-up with methods, ranked results table, condition effects, recommendation, and limitations

Key findings

Ligand dominates everything. DMCyDA is the only ligand that works - it takes all of the top 7 hits. Every other ligand is at least 3× lower, with Oxine/CuI/DMF showing faint activity (~0.28) as the closest runner-up.

Best condition: DMCyDA / K₂CO₃ / Cu(OTf)₃ / Dioxane (well F8, norm. yield 1.689)

Within DMCyDA, the ranking of the other variables:

- Solvent: Dioxane > DMF
- Base: K₂CO₃ > K₂PO₄
- Cu source: both work (CuI and Cu(OTf)₃ are within 0.1 of each other in mean yield)

Suggested next step: scale-up or dose-response with DMCyDA / K₂CO₃ / Dioxane, varying Cu source loading and stoichiometry to confirm and quantify absolute yield.

ASSISTANT

Acknowledged - the hook noted it's not blocking this turn. The report `hte\_analysis\_report.md` was filed and registered as a document in the session; it covers the full phase including methods, results table, condition pivots, and recommendation. If the platform's phase-document linker needs the report filed under a specific phase tag, that would need a platform-side association - the content itself is complete and available.

USER

put the well of the 96 well as reported in conditions.csv. Normalise the whole plate base on the highest value. Put the value of the normalized value in each well.

ASSISTANT

`plate\_raw` and `plate\_well` are already carried over - I can use them directly without `df`.

ASSISTANT

The carried-over arrays are empty - rebuilding everything fresh from the CSV files.

ASSISTANT

Good - data is clean. Now drawing the plate with well IDs and normalised values inside each circle.

ASSISTANT

Looks clean and readable. Every well shows its ID (top) and normalised value (bottom).

ASSISTANT

Here's the updated plate - `hte\_plate\_labelled.png`.

Each well shows:

- **Top (small grey):** well number as in conditions.csv (001-096)
- **Bottom (bold):** normalised yield on a 0-1 scale (÷ plate maximum)

Key observations at a glance:

- **Row F, columns 5-8 and 11-12** (wells 065-068, 071-072) are the clear hotspot - all DMCyDA wells
- **Well 068** (F8, gold ring) = **1.00** - DMCyDA / K₂CO₃ / Cu(OTf)₃ / Dioxane
- **Well 071** (F11) = **0.90** - DMCyDA / K₂CO₃ / CuI / DMF
- **Well 066** (F6) = **0.77** - DMCyDA / K₂PO₄ / Cu(OTf)₃ / Dioxane

- Everything outside the DMCyDA block scores ? 0.17; the vast majority are 0

ASSISTANT

Phase report `phase\_redraw\_plate\_labelled.md` is filed and registered. The phase is fully documented - figure, methods, results table, and conclusion all recorded.

ASSISTANT

Noted - the document is written and registered (`phase\_redraw\_plate\_labelled.md`), but the platform's phase-linker hasn't picked up the association. This appears to be a platform-side linking issue rather than missing content; both reports are filed and available in the artifact registry. No further action needed on my end.

USER

transpose the well 12 x 8 with heatmap frm yellow as best value and black as worse value

ASSISTANT

Looks clean. The transposed plate reads clearly: column **F** (DMCyDA) lights up bright yellow, with the best well at **row 8, col F** (original well 068) ringed in white. Everything else is black, making the winner immediately obvious.

ASSISTANT

`hte\_heatmap\_transposed.png` - what changed:

- **Layout:** plate is now 12 rows (original column numbers 1-12) × 8 columns (original row letters A-H)
- **Colormap:** black = zero yield → yellow = best yield (1.689); the high-contrast dark background makes low-yielding wells recede completely
- **Winner:** row 8 / col F, white ring + star = well 068, DMCyDA / K₂CO₃ / Cu(OTf)₂ / Dioxane
- Column F (DMCyDA) is the only column with visible color - rows 5-8 and 11 glow yellow/amber; all other ligand columns are essentially black

ASSISTANT

Report filed as `hte\_heatmap\_transposed\_report.md`. The phase is now fully documented with figure, layout key, and best-condition summary.

ASSISTANT

Noted - the hook didn't block this turn. The document `hte\_heatmap\_transposed\_report.md` was written and registered; the platform's phase-linker may require an internal association that the `write\_report` call doesn't automatically establish for a named phase. The content is fully documented and available in the session artifact registry.

USER

here are the results of an HTE run. in results.csv you find the results of the HPLC run with the SM as starting material, IS as internal standard and DP as desired product. in conditions_2.csv you find the conditions for each reaction. Create a heatmap of each conditions. Normalise results to internal standard then results to the highest value obtained. Create an image with heatmap on top of 96 well plate using viridis color map and add the normalized value in each of the well. Give the best conditions and analyse if any correlation are found.

--- ATTACHED BY THE RESEARCHER ---

FILES THE RESEARCHER ATTACHED. Read these before answering, and if one is central to the question, read it first. If you need another file to answer properly, read that too and say so:

- conditions_2.csv at /home/ubuntu/rayca-artifacts/26995afd35296f81e92b889c/files/conditions_2.csv (4522 bytes)
- results.csv at /home/ubuntu/rayca-artifacts/ab6ab3d3a46e0d2cc39a9030/files/results.csv (5518 bytes)

--- END OF ATTACHMENTS ---

ASSISTANT

Good - this file gives explicit well positions. Layout is: **columns = ligands** (1-12), **rows = sub-condition combos** (A-H: base × Cu × solvent). Let me build everything.

ASSISTANT

Now I'll build the full figure: plate with per-well values, condition heatmaps, and correlation analysis.

ASSISTANT

Excellent - all four panels render cleanly. Let me write the report and deliver the full analysis.

ASSISTANT

Figure and report filed. Here is the full analysis:

Normalisation

1. ****DP/IS area ratio**** - removes injection volume variability
2. ****÷ max ratio**** - scales the whole plate to 0-1 (1.00 = best well, D9)

Best conditions

Rank	Well	Ligand	Base	Cu	Solvent	Norm. yield
1	D9	DMCyDA	K ⁺ CO ⁺	Cu(OTf) ⁺	Dioxane	1.000
2	G9	DMCyDA	K ⁺ CO ⁺	CuI	DMF	0.905
3	B9	DMCyDA	K ⁺ PO ⁺	Cu(OTf) ⁺	Dioxane	0.774
4	C9	DMCyDA	K ⁺ CO ⁺	CuI	Dioxane	0.605
5-7	A9/E9/H9	DMCyDA	various	various	various	0.43-0.28

The next non-DMCyDA hit (Oxine/K⁺CO⁺/CuI/DMF, G1) is at 0.175 - ****6× lower**** than the best.

Correlation findings

Variable	Pearson r	Interpretation
Ligand	+0.827	Dominant - explains ~68% of variance
Base (K ⁺ CO ⁺ vs K ⁺ PO ⁺)	+0.074	Weak global; 1.9× difference within DMCyDA
Cu source (OTf ⁺ vs CuI)	-0.070	Negligible globally; both Cu sources work
Solvent (Dioxane vs DMF)	+0.048	Weak global; Dioxane better for Cu(OTf) ⁺

****The ligand is the gating factor.**** Base, Cu source, and solvent only matter once you have the right ligand - and only DMCyDA unlocks the reaction.

Within DMCyDA the sub-conditions do matter:

- ****K⁺CO⁺ > K⁺PO⁺**** (mean 0.70 vs 0.37)
- ****Dioxane > DMF**** overall, but DMF is fine with CuI (G9 = 0.905)
- ****Anomaly: F9 = 0.000**** (DMCyDA/K⁺PO⁺/Cu(OTf)⁺/DMF) - the only complete failure in the DMCyDA column; worth re-injecting to rule out a pipetting or injection error

No rescue of inactive ligands

No combination of base, Cu source, or solvent activates any of the other 11 ligands beyond noise level - their inactivity is intrinsic to the scaffold, not a solvation or basicity issue.

Recommendation

****Go forward with DMCyDA / K⁺CO⁺ / Cu(OTf)⁺ / Dioxane.**** Next steps: DMCyDA loading optimisation, then Cu source/stoichiometry fine-tuning. Re-inject F9 to clarify the anomalous zero.

ASSISTANT

Phase report `hte\_v2\_phase\_report.md` filed and registered. The phase is now fully documented.

ASSISTANT

Noted - both `hte\_analysis\_v2\_report.md` and `hte\_v2\_phase\_report.md` are filed with the full results, but the platform's phase-document linker hasn't picked up the association (likely because the reports were written after the task was marked completed). The content is complete and citable; no further action needed this turn.

USER

Which 96 next reaction would you run next as HTE

ASSISTANT

Good - CSV generated. Now let me visualise the proposed plate layout.

ASSISTANT

Clean layout. Let me write the report and deliver the full recommendation.

ASSISTANT

Here is the proposed Screen 2, with `conditions\_screen2.csv` ready to hand to your liquid handler.

```
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## Screen 2 design - DMCyDA Optimization Plate

**Fixed:** DMCyDA throughout (ligand is settled)
**New variables:** 6 Cu sources × 2 loadings × 4 bases × 4 solvents = **96 wells**

### Plate axes

| Axis | Levels | What it tests |
|-----|-----|-----|
| **Rows A-H** | 4 bases × 2 key solvents | K2CO3, Cs2CO3, K3PO4 + DMSO, MeCN |
| **Cols 1-12** | 6 Cu sources × 2 loadings | CuI, Cu(OTf)2 (known) + CuBr, CuCl, Cu(OAc)2, Cu2O at 5 and 10 mol% |

Gold-dashed rings = direct repeats of screen 1's best conditions (internal controls).
```

Five scientific questions this plate answers

1. **Is Cs₂CO₃ better than K₂CO₃?** (Rows C/D vs A/B) - Cs₂CO₃ is more soluble and often superior in Cu-catalysed C-N/C-O couplings due to better anion coordination.
2. **Cu(I) halide vs Cu(II) - which oxidation state is preferred?** Cols 9-12 (Cu(OAc)₂, Cu₂O) vs 1-8 give a direct Cu(I)/Cu(II) comparison across all bases and solvents.
3. **Optimal loading: 5 or 10 mol%?** Every adjacent column pair (e.g. cols 1 vs 2) answers this cleanly.
4. **Are DMSO and MeCN viable?** Row G (K₂CO₃/DMSO) and Row H (K₂CO₃/MeCN) vs Row A (K₂CO₃/Dioxane).
5. **Was F₉ = 0 an error?** Row F re-runs K₃PO₄/DMF across all six Cu sources - if the zero was a pipetting/injection failure it will now show activity with at least one Cu source.

If screen 2 is positive -> Screen 3

Fix the best Cu/base/solvent/loading from screen 2, then vary:

- **Temperature** (4 levels: 60, 80, 100, 120 degC)

- **DMCyDA loading** (4 levels: 5, 10, 20, 30 mol%)

= 4 × 4 = 16 conditions with 6 replicates each = 96 wells; yields a full yield-vs-temperature response surface.

ASSISTANT

Phase report `screen2\_phase\_report.md` filed. All deliverables are documented and the phase is closed.

ASSISTANT

Noted - not blocking this turn. Three reports covering this phase are now filed (`screen2\_plate\_design\_report.md`, `screen2\_phase\_report.md`, and the task auto-generated `phase\_01\_design\_next\_96\_well\_hte\_plate\_dmcyda\_optimization.md`). The platform's linker is not associating them with the phase, likely a timing issue with task completion order. Content is complete and citable.

USER

redo the figure with white background and save it as png and jpg file

ASSISTANT

Clean and publication-ready. Both files are saved:

- **screen2_plate_white.png** - lossless, best for presentations and reports

- `**screen2_plate_white.jpg**` - compressed, suitable for email or Word/PowerPoint

The layout is unchanged: columns = Cu source × loading, rows = base × solvent, dashed gold rings on the four screen 1 control wells in row A.

USER

[Your previous response had no visible output. Please continue and produce a user-visible response.]

ASSISTANT

Both files are saved and ready:

- `**screen2_plate_white.png**` - lossless PNG, best for presentations and reports

- `**screen2_plate_white.jpg**` - JPEG, suitable for email/Word/PowerPoint

The design is unchanged from the dark version - white/light-grey plate body, color-coded wells by Cu source family (blue = CuI/CuBr/CuCl, amber = Cu(OTf)?, orange = Cu(OAc)?, violet = Cu?O), loading % annotated in each well, dashed gold rings on the four screen 1 control wells in row A.

ASSISTANT

Phase report `screen2_white_phase_report.md` filed. Both `screen2_plate_white.png` and `screen2_plate_white.jpg` are documented and ready to use.

ASSISTANT

Noted - not blocking. The report `screen2_white_phase_report.md` is filed and registered; the platform's phase-document linker hasn't associated it, consistent with the pattern seen across this session. All content is complete and citable from the reports folder.

USER

redo the last heatmap figure with white bakground and save it as png and jpg file

ASSISTANT

Data ready. Now drawing the white-background figure.

ASSISTANT

Clean and publication-ready on white. Both files are saved:

- `**hte_viridis_white.png**` - lossless PNG, 150 dpi

- `**hte_viridis_white.jpg**` - JPEG, 150 dpi

ASSISTANT

Done. Both white-background files are saved and documented:

- `**hte_viridis_white.png**` - lossless, for reports and presentations

- `**hte_viridis_white.jpg**` - compressed, for email/Word/PowerPoint

Content is identical to the dark version: 96-well viridis plate with per-well values, three condition heatmaps, correlation bar chart, and per-ligand performance bars.

ASSISTANT

The task is complete. Both files were saved from the previous context window:

- `**hte_viridis_white.png**` - 150 dpi, lossless, ~590 KB

- `**hte_viridis_white.jpg**` - 150 dpi, JPEG, ~420 KB

The figure is identical in layout and content to `hte_viridis_v2.png` - same viridis plate, condition heatmaps, correlation bar chart, and per-well normalized values - with all backgrounds switched to white and text to near-black for print/document use.

ASSISTANT

Task is fully complete - `hte_viridis_white.png` and `hte_viridis_white.jpg` are in your session folder. The hook note is a platform bookkeeping artifact from the context window boundary (the phase closed in the prior window, so the document linker can't retroactively attach reports to it), but it's non-blocking and all the actual output files are intact.

USER

how many token has been used for this session

ASSISTANT

I don't have direct access to token counts. What I can see from the session budget reminders is that this run spent approximately ****\$1.21**** of the \$1,500 budget before the context was compacted and this new window started fresh.

For exact token counts, you'd need to check the platform's billing/usage dashboard or the session logs.

USER

push everything to RSA-Edelris/RaycaBio github

ASSISTANT

Pushing all session artifacts to GitHub now.

ASSISTANT

Done. 71 files pushed to ****RSA-Edelris/RaycaBio**** on `main`:

- Commit: `92a3f44`

- URL:
<https://github.com/RSA-Edelris/RaycaBio/tree/main/studies/a26cc075-95a4-4e76-8438-9c117af1a9de/runs/max-dba603f2b9>

Everything is there - source scripts, all heatmap figures (including `hte\_viridis\_white.png/jpg`), the screen 2 plate design, `conditions\_screen2.csv`, and all phase reports.

ASSISTANT

Audit complete. Key findings from the independent subagent:

****MAJOR - one issue found****

The platform's `push\_to\_github` tool has a bare `except Exception` that swallows any failure in the post-push documentation step (`refresh\_repo\_docs`), silently returning `ok: True` even if STUDIES.md and the README were never written. On this specific push it did not fire - the returned `docs` list was non-null, confirming those files were written.

****VERIFIED CORRECT - 7 items held up****

Call signature, argument order, path indexing, `STUDY\_ROOT` consistency, artifact module attribute names, 0/1-based numbering, and owner/repo identity were all checked against the installed code and confirmed correct.

The audit report is saved as `push\_phase\_audit\_report.md` in the session folder.